

Weiyan Xie (Vayne)

计算机科学博士生 · 香港科技大学 (HKUST)

wxieai@connect.ust.hk · +852 93643921/+86 19896595201 · 微信: Vayne_genius

· 个人主页: <https://vaynexie.github.io/me/>

经历

2025–2026 研究实习生, 华为香港 AI 框架与数据技术实验室

2022.09–2026.10 博士 (计算机科学与工程), 香港科技大学

- 华为博士奖学金获得者
- 导师: Prof. Nevin L. Zhang

2019.09–2020.12 硕士 (大数据技术), 香港科技大学

- 绩点: 4.11/4.3 (排名: 5/120)

2015.09–2019.06 学士 (统计学), 北师大香港浸会大学

- 绩点: 3.51/4.0 (排名: 3/70)
- 香港浸会大学一级荣誉毕业

研究陈述

我的研究聚焦于深度视觉与视觉-语言模型在真实场景中的应用, 重点关注可解释性、泛化能力、多模态大语言模型 (MLLMs) 的计算效率, 以及图像编辑中的可控性。我致力于构建诊断工具来分析模型当前依赖的关键信号, 并设计有针对性的机制, 引导模型学习更具因果相关性、更可信且更高效的行为。

Research Interests & Selected Work

Organized by research theme, each with representative publications and brief contributions.

主题 1: Vision-Language Models, MLLMs, and Agentic Visual Perception

Focus: Multimodal LLMs may rely on language priors rather than pertinent visual evidence, especially on long documents. I explore agentic perception frameworks that gather evidence iteratively to improve accuracy and efficiency.

- Representative work:
 - Kaican Li*, Weiyan Xie*, Lewei Yao, Jiannan Wu, Lanqing Hong, Yongxiang Huang, Nevin L. Zhang. **InSight-doc: Agentic Visual Perception for Long-Document Understanding**. 投稿中, 2026. * Equal contribution (alphabetical order)
 - Replaces fixed-resolution, single-pass pipelines with iterative perception that selectively acquires high-resolution crops on demand, advancing the accuracy–efficiency Pareto frontier.

主题 2: Controllable Image Editing and Generation

Focus: Controllable editing requires precise spatial and semantic guidance without costly retraining. I develop training-free methods that combine structural control with flexible prompt guidance.

- Representative work:

- **Weiyan Xie, Han Gao, Didan Deng, Kaican Li, April Hua Liu, Yongxiang Huang, Nevin L. Zhang. CannyEdit: Selective Canny Control and Dual-Prompt Guidance for Training-Free Image Editing. IEEE Conference on Artificial Intelligence (CAI), 2026.**
 - Enables selective edge-based structural control with dual-prompt guidance for training-free, controllable image editing.

主题 3: Robustness, Domain Generalization, and Adaptation

Focus: Foundation models can lose robustness during fine-tuning and fail under distribution shift. I study causal mechanisms behind domain generalization and design training objectives that anchor decisions to invariant, generalizable features.

- Representative work:
 - **Han Gao*, Kaican Li*, Weiyan Xie*, Zhi Lin, Yongxiang Huang, Luning Wang, Caleb Chen Cao, Nevin L. Zhang. Consistency Regularization for Domain Generalization with Logit Attribution Matching. Conference on Uncertainty in Artificial Intelligence (UAI), 2024. * Equal contribution (alphabetical order)**
 - Logit Attribution Matching (LAM) anchors predictions to domain-invariant causal features by matching logit attributions across semantic-sharing pairs.
 - **Kaican Li*, Weiyan Xie*, Yongxiang Huang, Didan Deng, Lanqing Hong, Zhenguo Li, Ricardo Silva, Nevin L. Zhang. Dual Risk Minimization: Towards Next-Level Robustness in Fine-Tuning Zero-Shot Models. Advances in Neural Information Processing Systems (NeurIPS), 2024. * Equal contribution (alphabetical order)**
 - Combats robustness vanishing during foundation-model fine-tuning by jointly optimizing empirical risk with worst-case risk estimated via CLIP and LLM-generated visual descriptions.

主题 4: Trustworthy and Explainable AI (XAI)

Focus: Deep classifiers often rely on spurious correlations rather than causally relevant visual evidence. My work develops explanation methods that diagnose these misaligned dependencies and surface discriminative, human-interpretable rationales.

- Representative work:
 - **Weiyan Xie, Xiao-Hui Li, Caleb Chen Cao, Nevin L. Zhang. ViT-CX: Causal Explanation of Vision Transformers. International Joint Conference on Artificial Intelligence (IJCAI), 2023.**
 - Estimates the causal effect of semantic patches on Vision Transformer predictions, moving beyond correlational saliency maps.
 - **Weiyan Xie, Xiao-Hui Li, Zhi Lin, Leonard K. M. Poon, Caleb Chen Cao, Nevin L. Zhang. Two-Stage Holistic and Contrastive Explanation of Image Classification. Conference on Uncertainty in Artificial Intelligence (UAI), 2023.**
 - Introduces Contrastive Whole-Output Explanation (CWOX), which explains a model's top-K labels by systematically contrasting visually confusable competitors.
 - **Nevin L. Zhang, Weiyan Xie, Zhi Lin, Guanfang Dong, Xiao-Hui Li, Caleb Chen Cao, Yunpeng Wang. Example Perplexity. arXiv:2203.08813, 2022.**
 - Proposes a diagnostic measure for assessing how well a model captures the structure of training examples.

教学经历

助教, Postgraduate Machine Learning (MSBD5012 / CSIT5910 / COMP5212): 2022 Fall, 2023 Spring, 2023 Fall, 2024 Fall, 2025 Fall

助教, COMP2011 Programming with C++ (UG core course): 2024 Spring

Co-supervised 15+ Master's independent projects: Most students received A-level grades

学术服务

会议审稿人 / 程序委员会成员

- NeurIPS (2023–2026)
- ICML (2023–2026)
- ICLR (2024–2026)
- UAI (2024–2026)
- AAAI (2025–2026)

荣誉与奖项

- 华为博士奖学金 (HKUST)
- 大数据技术硕士优秀学生奖 (HKUST), 2020
- 工学院优秀学生奖学金 (HKUST), 2020
- 学院一等奖学金 (BNBU), 2016–2019

技术技能

- ML & AI: LLMs and multimodal LLMs (VLM/MLLM), LLM/MLLM fine-tuning (SFT, instruction tuning), Reinforcement learning for LLMs (RLHF/RLAIF, GRPO), Flow-GRPO and RL-based post-training, Parameter-efficient tuning (LoRA/QLoRA, adapters), Inference optimization and serving (vLLM, quantization, batching), Agentic AI systems and tool-use workflows, RAG pipelines and retrieval-augmented generation, Model evaluation, benchmarking, and safety checks, Computer vision and explainable AI, Domain generalization and test-time adaptation
- 编程与框架: Python, C++, PyTorch, Transformers, DeepSpeed, Accelerate, Hugging Face, CUDA basics, TensorFlow/Keras, R, MATLAB
- 工具与平台: Linux, Git, Docker, Kubernetes basics, Ray, veRL, Weights & Biases, MLflow, FastAPI, REST API design, Vector databases, Prompt engineering, Code agents, Cursor, LaTeX, Spark, MySQL, Tableau, D3.js